

Why Distributed Solar Power Is Crucial

Sunlight is omnipresent. Why then centralise such well-distributed energy into a multi-megawatt solar farm, and redistribute it, losing 25 per cent in transmission?



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Distributed solar power generation is the opposite of centralised generation. In centralised generation, electricity is generated at central power plants that are sometimes located hundreds of kilometres away from the load centre where the consumers are located.

Oil, gas and coal are not present everywhere, so it makes sense to have a thermal power plant close to its fuel source. A hydro-electric power plant needs to be situated next to a waterfall and nuclear fission needs a protected zone around it. There is no option but to have these plants centralised.

Central power plants essentially need extra-high-tension (EHT) power

transmission. As a result, these involve a high investment cost and suffer huge losses in transmission. Fossil-fuel power plants, such as coal-fired power plants, also add heavily to pollution. In fact, they are responsible for as much as 40 per cent of the earth's greenhouse gas emissions.

Distributed power generation, small, is a method of generating electricity and other forms of energy from locally available energy sources such as sunlight, wind and biomass. These energy sources are omnipresent, and can be tapped at the point of actual use. They can meet the power needs of remote areas with lowest investment and also help to improve the quality of power in small cities and towns.

Distributed solar and wind energy generation can eliminate the need for inefficient 'central' solar and wind farms that are typically owned and operated by large electric utility companies. Small solar power plant can be locally owned by qualified small or medium enterprises (SMEs). These will also help to create much needed local employment.

Distributed power plants can cater well to the electrical power needs of a cluster of villages, ensuring power generation at the point of consumption, with zero transmission losses. Western countries, in fact, are planning for power self-sufficient communities based on locally available renewable energy sources.

Distributed power projects directly benefit the consumers and it makes great sense to set up these even where power grid has not reached.

If the grid is available, small power plants with 5-300kW capacity can feed the energy generated at a low voltage into a local substation. This locally generated power will first feed the local loads and the excess will go elsewhere, ensuring use of full power generation capacity of the plant. Distributed power plants also have very little greenhouse gas emissions.

Locally available renewable energy projects can deliver in just a couple of months compared to two to three years typically associated with larger-scale developments, and probably at a much lesser cost.

Why then centralise such well-distributed energy into a multi-megawatt solar or wind farm, and redistribute it, losing 25 per cent in transmission? Further, why use large and expensive